# Triangle Inequality Theorem; Converse of the Pythagorean Theorem

*Question Bank - Additional 6.01*

Name_____
Date_____
Period_____

*Show your work. Circle your final answer.*

**For questions 1 through 3, can the three lengths make a triangle?**

**1.** 4 m, 10 m, 12 m

**2.** 6 in, 17 in, 9 in

**3.** 14 ft, 7 ft, 7 ft

**For questions 4 and 5, two sides are given. What lengths can the third side be?**

**4.** 6 cm and 9 cm

**5.** 1 in and 8 in; the third side is a whole number of inches

**For questions 6 through 8, do the lengths make a right triangle?**

**6.** 2.4 m, 3.2 m, 4 m

**7.** 39 in, 36 in, 15 in

**8.** 8 m, 9 m, 12 m

**Choose one.**

**9.** Which set of lengths makes a right triangle?

- ○ A. 5, 6, 7
- ○ B. 14, 48, 50
- ○ C. 9, 12, 16
- ○ D. 3, 4, 7